

Churn Risk and Retention Report

Validated customer churn scoring from public Telco churn data

Rows analyzed

7,043

Validated customer rows

Observed churn

26.5%

Historical target rate

High risk

456

Priority accounts

Top 20% capture

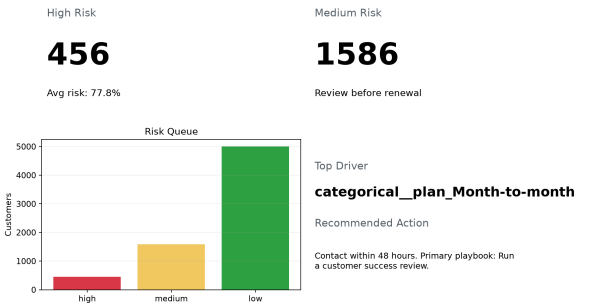
49.2%

Holdout churn capture

Executive Read

The model ranks customers by churn risk and turns that score into a retention queue. The highest-risk group should be reviewed first because the top 10% captures 28.2% of observed churners on the holdout split. This report separates model evidence from the full customer action file. Metrics, lift, calibration, and confidence intervals come from holdout customers; the scored queue covers the full customer base.

Churn Retention Executive Snapshot

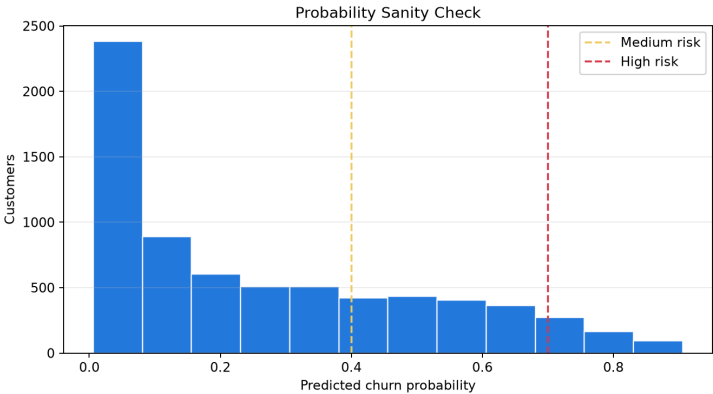


Dataset Summary

Metric	Value
rows	7043
columns	11
observed_churn_rate	26.54%
duplicate_customer_ids	0
leakage_warnings	0

Model Evidence

Metric	Value
roc_auc	0.8436
pr_auc	0.6562
precision	0.6512
recall	0.4991
f1	0.5651
brier_score	0.1361



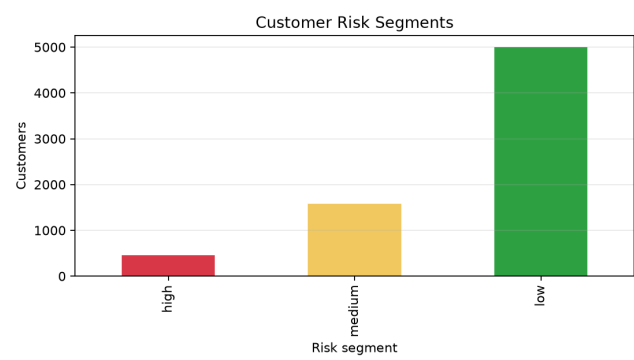
Lift and Calibration

decile	customers	churners	churn_rate	avg_prob	capture
1	212	158	0.7453	0.7300	0.2816
2	211	118	0.5592	0.5686	0.4920
3	211	98	0.4645	0.4531	0.6667
4	211	73	0.3460	0.3397	0.7968
5	212	41	0.1934	0.2345	0.8699

metric	estimate	ci_low	ci_high	samples
roc_auc	0.8436	0.8267	0.8620	300
pr_auc	0.6562	0.6113	0.7003	300

bin	customers	mean_pred	observed_rate	gap
1	212	0.7300	0.7453	0.0153
2	211	0.5686	0.5592	0.0093
3	211	0.4531	0.4645	0.0114
4	211	0.3397	0.3460	0.0063
5	211	0.2348	0.1943	0.0405

Risk Queue



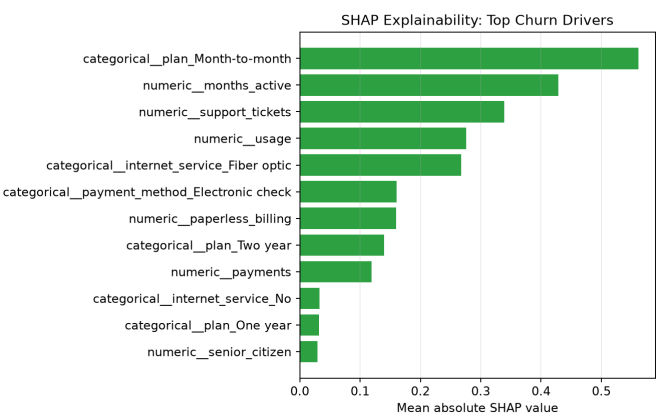
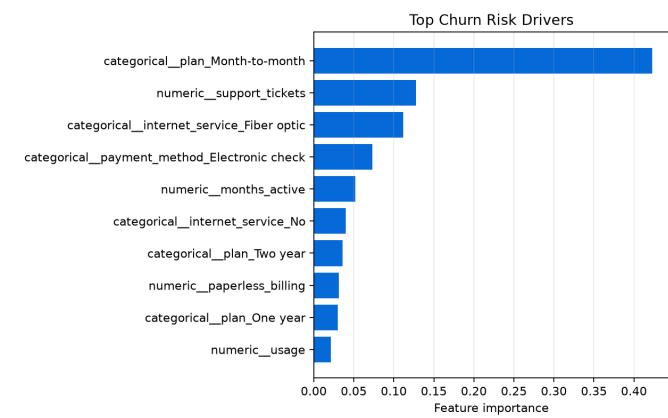
segment	customers	avg_prob
high	456	0.7777
medium	1586	0.5422
low	5001	0.1305

The PDF keeps actions short. Full recommended_action text is delivered in retention_recommendations.csv.

Top Recommended Accounts

customer_id	segment	risk	drivers	priority
7216-EWTRS	high	0.9058	general_engagement	1
9497-QCMMS	high	0.8941	general_engagement	1
9300-AGZNL	high	0.8941	general_engagement	1
5178-LMXOP	high	0.8923	general_engagement	1
8775-LHDJH	high	0.8866	general_engagement	1

Driver Evidence



Scope

This is a one-time churn analysis package from a supplied CSV. It does not claim guaranteed accuracy, production CRM integration, scheduled retraining, or hosted monitoring. The useful output is the reproducible evidence bundle plus the customer-level retention queue.